MIL-C-50245A (PA)
5 November 1975
Superseding
MIL-C-50245 (MU)
27 January 1969

MILITARY SPECIFICATION

CLIP, SAFETY, FOR HAND GRENADES

*This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers safety clips for use on the M61, M62 and MK3A2 grenades.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-26 - Bag, Plastic, Polyethylene, General Purpose

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC: 1330

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DRAWINGS

PICATINNY ARSENAL

9219914 - Clip, Safety
9244148 - Clip, Safety
9221580 - Pull Test Fixture

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

*(For meaning of asterisk see paragraph 6.5).

3. REQUIREMENTS

3.1 Materials. - The materials shall be in accordance with the applicable drawings and specifications.

3.2 Clip, safety. - The safety clips shall comply with all the requirements specified on Drawing (dwg.) 9219914 or dwg. 9244148 as applicable, and with all requirements specified in applicable specifications.

3.3 Release resistance. - The safety clip shall not be disengaged from its assembled position by a load of twenty-five (25) ounces, but shall be completely disengaged from the assembled position over the lever by a load of eighty (80) ounces.

3.4 Embrittlement. - The safety clip shall not break when pulled to the minimum point of permanent distortion.

3.5 Workmanship. - All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner, and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

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3.6 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

*4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the following items in sample quantities as indicated.

*4.3.2 Inspections to be performed. - See MIL-A-48078 and paras 4.4.2.1 thru 4.4.2.2, testing per para. 4.4.3, and Table I specified herein.

*4.3.3 Rejection. - See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-C-50245A (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Clip, Safety, For Hand Grenades				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Clip, Safety (Dwg. 9219914) Examination for Defects	25	100%	3.2	4.4.2.1
	Clip, Safety (Dwg. 9244148) Examination for Defects	25	100%	3.2	4.4.2.2
	Clip, Safety (Dwg. 9219914) Release Resistance	100	100%	3.3	4.5.1/Dwg. 9221580
	Embrittlement	32	100%	3.4	4.5.2
	Clip, Safety (Dwg. 9244148) Release Resistance	100	100%	3.3	4.5.1/Dwg. 9221580
	Embrittlement	32	100%	3.4	4.5.2

NOTES:

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4.4 Quality conformance inspection.

*4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

*4.4.2 Examination. - (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50245A (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Clip, Safety		9219914
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
Test	Release resistance	500	(a)	3.3	4.5.1
Major B					
131	Profile incorrect		0.40%	3.2	Gage
132	Location of formed loop		0.40%	3.2	Gage
133	Diameter of wire		0.40%	3.2	Gage
Minor					
201	Width of formed rectangle		0.65%	3.2	Gage
202	Profile of formed loop		0.65%	3.2	Gage
203	Protective finish missing or contains bare spot		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES: (a) Accept with 5 or less defects
Reject with 6 or more defects

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-50245A (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Clip, Safety		9244148
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
Test	Release resistance	500	(a)	3.3	4.5.1
Major B					
131	Diameter of wire		0.40%	3.2	Gage
132	Diameter of loop		0.40%	3.2	Gage
133	Profile incorrect		0.40%	3.2	Gage
134	Angle between loop and formed rectangle leg		0.40%	3.2	Gage
Minor					
201	Width of formed rectangle		0.65%	3.2	Gage
202	Protective finish missing or contains bare spot		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

(a) Accept with 5 or less defects
Reject with 6 or more defects

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4.4.3 Testing

4.4.3.1 Embrittlement (see 3.4) - Major A defect. - Thirty-two (32) safety clips shall be selected from each lot for this test. The lot shall be rejected if any safety clip fails to comply with the requirement. Clips subjected to test shall not be returned to the lot.

4.4.4 Inspection Equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/ Inspection Method' column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (see 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and 6.2 herein.

4.5 Test methods and procedures

4.5.1 Release resistance. - The safety clip shall be assembled on the inspection equipment specified on dwg. 9221580 and in 4.4.4. The load shall be applied slowly and steadily in the clip releasing direction until the clip disengages from its assembled position. The applied load that caused disengagement shall be recorded for determination of lot disposition. The test samples may be returned to the lot.

4.5.2 Embrittlement. - The loop end and the formed rectangle of the safety clip shall be pulled in the opposite direction by any approved method. Clips so tested shall be scrapped.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 The safety clips shall be packaged, packed and marked in accordance with MIL-STD-1169 and the following additional instructions.

5.1.2 Level C.

5.1.2.1 Unit packaging. - One hundred (100) clips shall be nested on a paperboard tube with tube ends capped or taped

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to prevent clips from sliding off tube. Tubes shall be constructed of kraft or commercial chipboard spiral or convolute wound having an outside diameter of 3/4 inch and wall thickness of .015 inch minimum. The length of the tube shall be approximately ten (10) inches.

5.2 Packing

5.2.1 Level C. - Each layer of unit packages shall be separated by a plain fiberboard pad equal in size to the inside length and width of the shipping container less 1/16 inch. Additional pads shall be placed on top of the contents as required to provide a tight pack.

6. NOTE

6.1 Intended use. - This is used for the safety clips in M61 and M62 Grenades, Dwg. 9219914 and MK3A2 Grenades, Dwg. 9244148.

6.2 Ordering data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for submission of first article samples.

c. Provisions of MIL-A-48078.

6.3 Submission of Inspection Equipment Designs for Approval. - (See MIL-A-48078). Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T, Dover, New Jersey 07801.

6.4 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801.

Asterisks -

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations.

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Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1330-A-064

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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